Week-8
Mathematics for Data Science - 2
Orthogonality, Projection, Gram-Schmidt process, Orthogonal transformation
Practice Assignment Solution

1. **Note:** Suppose W_1 **and** W_2 are subspaces of a vector space V. Then sum of two subspaces is defined as:

$$W_1 + W_2 = \{w_1 + w_2 \mid w_1 \in W_1, w_2 \in W_2\}.$$

One can verify that $W_1 + W_2$ is a subspace of V.

1 Multiple Choice Questions (MCQ)

1. An inner product on $\mathbb{R}^3$ is defined as:

$$\langle ., . \rangle : \mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}$$
$$\langle (x_1, x_2, x_3), (y_1, y_2, y_3) \rangle = x_1y_1 + x_2y_2 + x_3y_3.$$

Match a set S in Column A with the set T given in column B, so that $T \subset Span(S)^{\perp}$. Match a set in column B with the property it satisfies in column C.

	Set (S) (Column A)		set (T) (Column B)		Properties (Column C)
a)	{(-2,1,3)}	i)	{(0,1,1),(2,-1,1)}	1)	forms an orthonormal basis of a vector space which is isomorphic to $\mathbb{R}^2$
b)	{(1,-2,1)}	ii)	{(1,-1,1)(2,1,1)}	2)	orthogonal but not orthonormal
c)	{(1,1,-1)}	iii)	$\{\frac{1}{\sqrt{2}}$ (0,-1,1) $\}$	3)	not orthogonal
d)	{(0,1,1)(1,1,1)}	iv)	$\{\frac{1}{\sqrt{3}}$(1,1,1),$\frac{1}{\sqrt{2}}$(-1,0,1)}	4)	orthonormal but does not form a basis of a vector space which is isomorphic to $\mathbb{R}^2$

Table : M2W9P1

Which of the following option are true?

- ◯ Option 1: d$\rightarrow$ ii $\rightarrow$ 3
- ◯ **Option 2:** d$\rightarrow$ iii $\rightarrow$ 4
- ◯ Option 3: c $\rightarrow$ ii $\rightarrow$ 4
- ◯ Option 4: a $\rightarrow$ iv $\rightarrow$ 1
- ◯ **Option 5:** a $\rightarrow$ ii $\rightarrow$ 3
- ◯ Option 6: b $\rightarrow$ iv $\rightarrow$ 2

- $S = \{(-2,1,3)\}$

$\langle (-2,1,3),(1,-1,1)\rangle = -2-1+3=0$

$\langle (-2,1,3),(2,1,1)\rangle = -4+1+3=0$

Therefore, $T=\{(1,-1,1),(2,1,1)\} \subset (\mathrm{Span}(S))^{\perp}$

$\langle (1,-1,1),(2,1,1)\rangle = 2-1+1=2\neq 0$

$\Rightarrow$ T- is not an orthogonal Set.

- $S=\{(1,-2,1)\}$

$\langle (1,-2,1), \frac{1}{\sqrt{3}}(1,1,1)\rangle = \frac{1}{\sqrt{3}}(1-2+1)=0.$

$\langle (1,-2,1), \frac{1}{\sqrt{2}}(-1,0,1)\rangle = \frac{1}{\sqrt{2}}(-1+0+1)=0.$

Therefore $T=\{\frac{1}{\sqrt{3}}(1,1,1), \frac{1}{\sqrt{2}}(-1,0,1)\} \subseteq (\mathrm{Span}(S))^{\perp}.$

$\langle \frac{1}{\sqrt{3}}(1,1,1), \frac{1}{\sqrt{2}}(-1,0,1)\rangle = \frac{1}{\sqrt{6}}(-1+0+1)=0$

$\Rightarrow$ T- is orthogonal Set.

$$\left\| \tfrac{1}{\sqrt{3}}(1,1,1) \right\| = \sqrt{\tfrac{1}{3}+\tfrac{1}{3}+\tfrac{1}{3}} = 1$$

$$\left\| \tfrac{1}{\sqrt{2}}(-1,0,1) \right\| = \sqrt{\tfrac{1}{2}+\tfrac{1}{2}} = 1$$

$\Rightarrow$ T-is a orthonormal set.

$V := \mathrm{Span}(T) \Rightarrow$ T-is a basis of V

& $\dim(V) = 2$.

$$T : \mathbb{R}^2 \longrightarrow V$$

$$(1,0) \longmapsto \tfrac{1}{\sqrt{3}}(1,1,1)$$

$$(0,1) \longmapsto \tfrac{1}{\sqrt{2}}(-1,0,1).$$

that is, $T(x,y) = \frac{x}{\sqrt{3}}(1,1,1) + \frac{y}{\sqrt{2}}(-1,0,1)$ and T gives an isomorphism between V and $\mathbb{R}^2$.

Similarly, you can try for the remaining options.

2. Suppose W_1 and W_2 are subspaces of a vector space V. Let P_{W_1} and P_{W_2} denote the projection from V to W_1 and V to W_2 respectively and consider the following statements:

- **Statement P:** If $P_{W_1} + P_{W_2}$ is a projection from V to $W_1 + W_2$, then $P_{W_1} \circ P_{W_2} + P_{W_2} \circ P_{W_1} = 0$.
- **Statement Q:** $P_{W_1}^2 := P_{W_1} \circ P_{W_1}$ is not a projection from V to W_1.
- **Statement R:** If A is the matrix representation of P_{W_2}, then A can not be a symmetric matrix.
- **Statement S:** $P_{W_1} - P_{W_2}$ is a projection from V to $W_1 + W_2$.

Find the number of correct statements [Ans: 1]

A linear transformation "T" is a projection

Iff $T \circ T = T$

Statement P: $P_{W_1} + P_{W_2}$ is a projection from V to $W_1 + W_2$

$\Leftrightarrow (P_{W_1} + P_{W_2}) \circ (P_{W_1} + P_{W_2}) = P_{W_1} + P_{W_2}$.

$\Leftrightarrow P_{W_1} \circ P_{W_1} + P_{W_1} \circ P_{W_2} + P_{W_2} \circ P_{W_1} + P_{W_2} \circ P_{W_2} = P_{W_1} + P_{W_2}$

$\Leftrightarrow P_{W_1} + P_{W_1} \circ P_{W_2} + P_{W_2} \circ P_{W_1} + P_{W_2} = P_{W_1} + P_{W_2}$

[It is given that P_{W_1} and P_{W_2} are projection maps

$\Rightarrow P_{W_1} \circ P_{W_1} = P_{W_1}$ and $P_{W_2} \circ P_{W_2} = P_{W_2}$].

$\Leftrightarrow P_{W_1} \circ P_{W_2} + P_{W_2} \circ P_{W_1} = 0$.

Statement Q: By definition, P_{W_1} is a Projection.

Statement R: $V = \mathbb{R}^2$, $W_1 = x$-axis, $W_2 = y$-axis

$$P_{W_2} : V \longrightarrow V$$
$$(x, y) \longmapsto (0, y)$$

Take $\mathcal{B} = \{(1,0), (0,1)\}$ as a basis for both domain and co-domain.

Matrix representation of P_{W_2} w.r.t $\mathcal{B}$ is

$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$, which is a symmetric matrix.

Statement: S

$P_{W_1} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ $\qquad$ $P_{W_2} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$

$P_{W_1}(x, y) = (x, 0)$ $\qquad$ $P_{W_2}(x, y) = (0, y)$

$$(P_{W_1} - P_{W_2}) \circ (P_{W_1} - P_{W_2})(1, 1) \neq (P_{W_1} - P_{W_2})(1, 1)$$

$\Rightarrow P_{W_1} - P_{W_2}$ is not a Projection.

3. Which of the followings option(s) is (are) true?

- ◯ Option 1: Let $u, v,$ and w be three vectors in $\mathbb{R}^2$ and $\langle .,. \rangle$ be an inner product on $\mathbb{R}^2$. If $\langle u, v \rangle = \langle u, w \rangle$, then $v = w$
- ◯ **Option 2:** Let V be an inner product space. If $\langle u, v \rangle = 0$ for all $u \in V$, then $v = 0$
- ◯ Option 3: Let V be an inner product space and $u, v \in V$. If $\langle u+v, u-v \rangle = 0$, then $\|u\| \neq \|v\|$
- ◯ **Option 4:** Let V be an inner product space. If $S \subset V$ then $S^{\perp} = (span(S))^{\perp}$
- ◯ **Option 5:** Suppose $\{u_1, u_2, \ldots u_m\}$ be an orthogonal set of vectors in an inner product space V, then $\|u_1 + u_2 + \ldots + u_m\|^2 = \|u_1\|^2 + \|u_2\|^2 + \ldots + \|u_m\|^2$.

Option 1: Take $V = \mathbb{R}^2$ and $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_1 + x_2 y_2$

$u = (1, -1)$, $v = (1, 1)$, $w = (0, 0)$.

$\langle u, v \rangle = \langle (1, -1), (1, 1) \rangle = 0 = \langle u, w \rangle$

but $v \neq w$.

Option 2: Given: $\langle u, v \rangle = 0$ for all $u \in V$.

In particular, It is true for $u = v$

$\Rightarrow \langle v, v \rangle = 0 \Rightarrow v = 0$.

Option 3: $\langle u+v, u-v \rangle = 0$.

$\Rightarrow \langle u, u \rangle + \langle u, -v \rangle + \langle v, u \rangle + \langle v, -v \rangle = 0$

$\Rightarrow \langle u,u\rangle - \langle u,v\rangle + \langle v,u\rangle - \langle v,v\rangle = 0$

$\Rightarrow \langle u,u\rangle - \langle v,v\rangle = 0$

$\Rightarrow \langle u,u\rangle = \langle v,v\rangle \Rightarrow \|u\|^2 = \|v\|^2.$

<u>Option 4:</u> Suppose $S = \{v_1, v_2, ---, v_n\}$

<u>Claim:</u> $S^\perp = (\mathrm{Span}(S))^\perp.$

Let $v \in S^\perp \Rightarrow \langle v_i, v\rangle = 0$ For $i = 1 \sim n$.

Take $x \in \mathrm{Span}(S) \Rightarrow x = a_1v_1 + a_2v_2 + - + a_nv_n.$

where $a_i \in \mathbb{R}$

Now $\langle x, v\rangle = \langle a_1v_1 + a_2v_2 + --- + a_nv_n, v\rangle$

$= \langle a_1v_1, v\rangle + \langle a_2v_2, v\rangle + --- + \langle a_nv_n, v\rangle$

$= a_1\langle v_1, v\rangle + a_2\langle v_2, v\rangle + -- + a_n\langle v_n, v\rangle$

$= 0$

<u>Option 5:</u> $\{u_1, u_2, ---, u_m\}$ orthogonal set of vectors, that is, $\langle u_i, u_j\rangle = 0$ if $i \neq j$.

$$\|u_1+u_2+\cdots+u_m\|^2 = \langle u_1+u_2+\cdots+u_m,\ u_1+u_2+\cdots+u_m\rangle$$

$$= \langle u_1, u_1+u_2+\cdots+u_m\rangle + \langle u_2, u_1+u_2+\cdots+u_m\rangle + \cdots$$

$$+ \langle u_m, u_1+u_2+\cdots+u_m\rangle$$

$$= \langle u_1, u_1\rangle + \langle u_2, u_2\rangle + \cdots + \langle u_m, u_m\rangle$$

$$= \|u_1\|^2 + \|u_2\|^2 + \cdots + \|u_m\|^2$$

4. Which of the following option(s) is/are true?

- ◯ **Option 1:** Let A and B be orthogonal matrices of order 3. Then A and B are equivalent matrices.
- ◯ Option 2: Let A and B be orthogonal matrices. Then A and B are similar matrices.
- ◯ **Option 3:** Let A be a rotation matrix corresponding to the anti-clock wise rotation of the XY-plane about the Z-axis with angle θ. Then nullity of the matrix A is 0.
- ◯ **Option 4:** Let A be a rotation matrix corresponding to the anti-clock wise rotation of the XY-plane about the Z-axis with angle θ and B be a rotation matrix corresponding to the anti- clock wise rotation of the YZ-plane about the X -axis with angle β. Then A and B are equivalent matrices.

Option 1: If A and B are orthogonal matrices of order 3 then $\text{Rank}(A) = \text{Rank}(B) = 3$. Therefore, A and B are equivalent.

Option-2. Take $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

Both are orthogonal but not similar.

Option-3:

$$A = \begin{pmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$\det(A) = 1 \Rightarrow A$ is invertible

$\Rightarrow$ Nullity of A is Zero.

Option-4:

$$A = \begin{pmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix} \qquad B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\beta) & -\sin(\beta) \\ 0 & \sin(\beta) & \cos(\beta) \end{pmatrix}.$$

$$\mathrm{Det}(A) = \mathrm{Det}(B) = 1 \Rightarrow \mathrm{Rank}(A) = \mathrm{Rank}(B) = 3$$

$\Rightarrow$ A and B are equivalent.

2 Numerical Answer Type

5. Let $u = (1, 2, 1)$ and $v = (1, 2, 3)$ be the vectors of the inner product space $\mathbb{R}^3$ with usual inner product. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be an orthogonal linear transformation. If θ is an angle between $T(u)$ and $T(v)$, then find the value of $||T(u)||||T(v)|| \cos\theta$. [Ans: 8]

- If θ is the angle between $T(u)$ and $T(v)$ then

$$\cos(\theta) = \frac{T(u) \cdot T(v)}{\|T(u)\| \, \|T(v)\|} \Rightarrow \|T(u)\| \, \|T(v)\| \cos\theta = T(u) \cdot T(v).$$

T-is orthogonal $\Rightarrow T(u) \cdot T(v) = u \cdot v = 1+4+3$
$= 8.$

6. Consider a set $W = \{(1,1,1)\}$ from the inner product space $\mathbb{R}^3$. Find the dimension of the subspace $W^\perp$, where $W^\perp$ is the collection of the vectors which are orthogonal to the vector (1,1,1). [Ans: 2]

- $W^\perp = \{(x, y, z) \mid \langle (x, y, z), (1, 1, 1) \rangle = 0\}$

$= \{(x, y, z) \in \mathbb{R}^3 \mid x + y + z = 0\}.$

$\text{Dim}(W^\perp) = 2.$

7. Let $v = (3, 1, 2)$ be a vector in $\mathbb{R}^3$. If (a, b, c) is the vector obtained from v after the anti-clock wise rotation of ZX-plane with angle $45°$ about the Y- axis, then find the value of $\sqrt{2}(a + b + c - 1)$. [Ans: 6]

$$\begin{pmatrix} \cos(45) & 0 & -\sin(45) \\ 0 & 1 & 0 \\ \sin(45) & 0 & \cos(45) \end{pmatrix} \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}.$$

$$\Rightarrow \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \end{pmatrix} \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} \\ 1 \\ 5/\sqrt{2} \end{pmatrix}$$

$$\sqrt{2}\,(a+b+c-1) = \sqrt{2} \times 6/\sqrt{2} = 6.$$

3 Comprehension Type Question

With a particular frame of reference (in $\mathbb{R}^2$) the position of Rahul's house is at the point (8,1). There are two roads Road 1 and Road 2 along the lines $y = 2x$ and $y = -2x$ respectively on the $XY-$plane.
Answer questions 8,9 and 10 using the given information.

8. Which of the following option(s) is (are) true?

   - ◯ **Option 1:** If Rahul wants to travel the minimum distance from his house to Road 1, then he will meet Road 1 at (2,4).
   - ◯ Option 2: If Rahul wants to travel the minimum distance from his house to Road 1, then he will meet Road 1 at $\frac{6}{5}(1, -2)$.
   - ◯ **Option 3:** If Rahul wants to travel the minimum distance from his house to Road 2, then he will meet Road 2 at $\frac{6}{5}(1, -2)$.
   - ◯ Option 4: If Rahul wants to travel the minimum distance from his house to Road 2, then he will meet Road 2 at (2,4).

The closest Point from Rahul's home to Road 1 is the orthogonal Projection of the Point (8, 1) on the line $Y=2x$.

The line $Y=2x$ is same as $V_1 = \{(x, 2x) \mid x \in \mathbb{R}\}$
$= \text{Span}\{(1,2)\}$

The Projection of (8, 1) onto V_1 is

$$\frac{1}{\langle (1,2),(1,2)\rangle} \langle (8,1),(1,2)\rangle (1,2) = \frac{10}{5}(1,2) = (2,4).$$

- Equation for road 2 is $y = -2x$, which is same as $V_2 = \{(x, -2x) \mid x \in \mathbb{R}\} = \text{Span}\{(1,-2)\}$

The projection of $(8,1)$ onto V_2 is

$$\frac{1}{\langle (1,-2),(1,-2)\rangle} \langle (8,1),(1,-2)\rangle (1,-2) = \frac{6}{5}(1,-2)$$

9. Let θ be the acute angle between Road 1 and Road 2, then what will be the value of $5\cos(\theta)$? [Answer: 3]

- If θ is an acute angle then $\cos(\theta)$ must be non-negative.

$$\cos(\theta) = \left| \frac{(1,2)\cdot(1,-2)}{\sqrt{5}\cdot\sqrt{5}} \right| = \frac{3}{5}$$

$$\Rightarrow 5\cos(\theta) = 3.$$

10. Let $\langle .,. \rangle$ denote the standard inner product on $\mathbb{R}^2$, i.e., $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_1 + x_2 y_2$. Suppose $W_1 = \{(x, y) \mid y = 2x\}$ and $W_2 = \{(x, y) \mid y = -2x\}$ are two subspaces of $\mathbb{R}^2$. Which of the following options is correct?

- ○ Option 1: $W_1^\perp = \{(x, y) \mid 2y = x\}$
- ○ Option 2: $W_2^\perp = \{(x, y) \mid 2y = -x\}$
- ○ **Option 3:** $W_1^\perp = \{(x, y) \mid 2y = -x\}$
- ○ **Option 4:** $W_2^\perp = \{(x, y) \mid 2y = x\}$

$$W_1 = \{(x,y) \mid y = 2x\} = \{(x, 2x) \mid x \in \mathbb{R}\}$$
$$= \text{Span}\{(1,2)\}$$

$$W_1^\perp = \{(x,y) \mid \langle (x,y), (1,2) \rangle = 0\}$$
$$= \{(x,y) \mid x + 2y = 0\}$$
$$= \{(x,y) \mid x = -2y\}$$

$$W_2 = \{(x,y) \mid y = -2x\} = \{(x, -2x) \mid x \in \mathbb{R}\}$$
$$= \text{Span}\{(1,-2)\}$$

$$W_2^\perp = \{(x,y) \mid \langle (x,y), (1,-2) \rangle = 0\}$$
$$= \{(x,y) \mid x - 2y = 0\}$$
$$= \{(x,y) \mid x = 2y\}$$